

ORIGINAL ARTICLE

Noninvasive diagnosis of oral squamous cell carcinoma by multi-level deep residual learning on optical coherence tomography images

Wei Yuan¹  | Jinsuo Yang¹ | Boya Yin¹ | Xingyu Fan¹ | Jing Yang¹ | Haibin Sun¹ | Yanbin Liu¹ | Ming Su¹ | Sen Li² | Xin Huang¹ 

¹Department of Oral and Maxillofacial-Head and Neck Oncology, Beijing Stomatological Hospital, School of Stomatology, Capital Medical University, Beijing, China

²College of Science, Harbin Institute of Technology, Shenzhen, China

Correspondence

Xin Huang, Department of Oral and Maxillofacial-Head and Neck Oncology, Beijing Stomatological Hospital, School of Stomatology, Capital Medical University, Beijing, China.
Email: huangxin@ccmu.edu.cn

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Abstract

Background: Oral Squamous Cell Carcinoma (OSCC) is one of the most severe cancers in the world, and its early detection is crucial for saving patients. There is an inevitable necessity to develop the automatic noninvasive OSCC diagnosis approach to identify the malignant tissues on Optical Coherence Tomography (OCT) images.

Methods: This study presents a novel Multi-Level Deep Residual Learning (MDRL) network to identify malignant and benign(normal) tissues from OCT images and trains the network in 460 OCT images captured from 37 patients. The diagnostic performances are compared with different methods in the image-level and the resected patch-level.

Results: The MDRL system achieves the excellent diagnostic performance, with 91.2% sensitivity, 83.6% specificity, 87.5% accuracy, 85.3% PPV, and 90.2% NPV in image-level, with 0.92 AUC value. Besides, it also implements 100% sensitivity, 86.7% specificity, 93.1% accuracy, 87.5% PPV, and 100% NPV in the resected patch-level.

Conclusion: The developed deep learning system expresses superior performance in noninvasive oral squamous cell carcinoma diagnosis, compared with traditional CNNs and a specialist.

KEYWORDS

artificial intelligence, automatic diagnosis, deep learning, noninvasive diagnosis, oral cancer, oral squamous cell carcinoma

1 | INTRODUCTION

Oral Squamous Cell Carcinoma (OSCC) is one of the most widespread cancers, with severe modality in the whole world (Cheung et al., 2021; Siegel et al., 2021). As we all know, it is diagnosed by the confirmation of biopsy and histopathology examination, which requires tissue slicing, staining, and specific molecular pathology, causing inevitable experience with painful, invasive, and time-consuming operations for patients (Costello, 2016; Ramos-Garcia et al., 2021). In recent years, Optical Coherence Tomography (OCT) has primarily become one of the advanced optical imaging technologies with the nondestructive observing capability to inspect the molecular composition of oral tissues (Huang et al., 1991). Moreover, OCT imaging

can precisely perceive the cancer surroundings without invasive, time-consuming operations, such as staining and slicing. Another significant advantage of OCT imaging is that the resolution keeps similar quality to histopathological inspection, thus containing sufficient information to represent the micron-scale structure in living oral tissues (Huang et al., 1991). Because high morphological resolution in OCT image provides much information to be fully exploited for disease analysis quantitatively, OCT imaging has been successfully applied in several disease diagnoses, e.g., ophthalmology (Abdelazeem et al., 2019; Costello, 2016), cardiology (Yonetsu et al., 2013), gastroenterology (Tsai et al., 2017), and dermatology (Adabi et al., 2018; Jerjes et al., 2020). Benefiting from the great efforts of researchers and engineers, OCT technology has been developed and has the

overwhelming potential to be applied to precisely identifying oral cancers in a noninvasive, real-time manner that will effectively decrease the pain of OSCC patients and assist the surgeons in determining the diagnosis of oral squamous cell carcinoma.

The crucial tasks hindering in OCT-based OSCC diagnostic so far are the feature characteristics exploitation from optical coherence tomography data and how to decide by complicated analysis. In clinical diagnosis, this professional procedure is time-consuming and cost-expensive when human specialists implement it. Therefore, the automatic oral tissue classification system and objective computerized evaluation tool are demanded to achieve the complex feature representation and accurate image analysis for OCT-based OSCC diagnosis. Such an OCT imaging analysis system can replace expensive human power to identify the malignant and benign oral tissues, making surgeons relaxed and paying more attention to the clinical treatment of oral squamous cell carcinoma. Inspired by the successful applications of Artificial Intelligence (AI) in cancer diagnosis (Echle et al., 2021; Munir et al., 2019; Tandel et al., 2019), AI makes it possible to establish the automatic OCT-based OSCC diagnosing system due to its excellent capacities of feature learning and logical reasoning.

As an advanced technology of AI, deep learning has developed into an outstanding feature representation tool such that it can interpret the tissue images into disease determination by a series of nonlinear transformation algorithms. To establish an AI-based OSCC diagnosis system in a noninvasive manner, this paper now presents a novel deep learning network for diagnosing oral squamous cell carcinoma on OCT images, that is, the Multi-Level Deep Residual Learning (MDRL) model. The MDRL system corporately extracts high-level and low-level semantic information from the residual neural network (ResNet) (He et al., 2016) with a spatial attention mechanism, achieving a fast and reliable classification of malignant and benign tumors from noninvasive OCT images. ResNet is a specific deep learning technique that has been the key model to improve classification performance in the last years significantly and has not been applied and adapted to oral OCT imaging so far. The key idea behind our deep learning approach (MDRL) is to find the right classification decision and the inherent multi-level representations at the same time. Consequently, subjective and manual design of features from the OCT images is avoided, which can easily result in a bias of the later classification step. The model training in this paper means to find those parameters, that is, weights, of the neural network that minimizes the difference between expected and predicted outputs of the classifier.

2 | MATERIALS AND METHODS

2.1 | Study design

The AI-based medical image analysis research for oral squamous cell carcinoma is conducted at the Department of Oral and Maxillofacial Head and Neck Oncology, Beijing Stomatological Hospital, School

of Stomatology, Capital Medical University, Beijing, China. Approval for this work was obtained through the local Institute Review Board (No. CMUSH-IRB-KJ-PJ-2021-09), and written informed consent was obtained from research participants.

This research involves 37 patients (Table 1) with primary oral squamous cell carcinoma treated from Sep 2020 to Jun 2021 in Beijing Stomatological Hospital, Capital Medical University, China. The inclusion criteria are age ≥ 18 years, diagnosed by the standard histopathology, and obtained informed consent. Each patient has resected 3–6 tumor tissues from the tongue, gingiva, cheek, mouth floor, and oropharynx, ranging from early to terminal stages. As for the tumor stages, 8.1% of patients are in stage I, 37.8% are in stage II, 13.5% are in stage III, and 40.6% are in stage IV. Besides, the patients are resected different patches(tissues), and each resected patch is captured into 3–6 OCT images from different views, and the statistical data of the resected tumor tissues are summarized in Table 2. Thus, this paper introduces the performance in two levels to measure the model, including the image-level and the resected patch-level.

2.2 | Implementation

From OCT imaging from 37 patients, we capture 468 3D images, consisting of 224 malignant and 244 benign data, all of which have corresponding histopathological diagnoses without repeating images. All images are captured by an ultra-high-resolution microscopy imaging system, with 300nm spectral bandwidth, 800nm center wavelength, and 1.5 μ m axial resolutions in tissue super-continuum fiber laser source. Each OCT image is manually marked by two senior dental specialists with professional diagnoses to justify the biopsy of benign mucosal tissues and divided into training (346), and testing (122) sets. As for the training and testing group

TABLE 1 Patient demographics

Patients	Age, mean	Sex, F/M	Patch, mean
37	58.2 $\pm$ 12.4	14/23	3.49 $\pm$ 0.78

Note: Data are numbers of patients, mean age, different sexes, and mean resected patches.

Abbreviations: F, female; M, male; N, Number.

TABLE 2 Statistics of resected regions

Region	Training	Testing
Tongue	44	15
Gingiva	25	8
Cheek	21	2
Mouth floor	8	3
Oropharynx	2	1

Note: Totally, 100 resected patches are chosen as the training set, and 29 resected patches are tested.

division, we randomly separate the dataset to form the training and testing sets to ensure no train-test data contamination in the resected patch-level. The proposed deep learning (AI) system is implemented by Pytorch on Nvidia Geforce 2080Ti, optimized by training data, and evaluated on the testing data. Before being analyzed, each 3D OCT image is first released into around 100 2D sub-images, and each 3D image from the training set is preprocessed by random crops and resized into 224×224 before being fed into the AI system. As for the time-consuming of our model, it costs 7.3 ms for training each image and spends 0.11 ms for testing one. To make the model in this paper reproducible, we released our source code and model's parameters on Github (<https://github.com/yuanweiccmu/MDRL>).

2.3 | The developed convolutional neural network

To establish a deep learning-based AI system for diagnosing Oral Squamous Cell Carcinoma (OSCC) from noninvasive OCT images, we develop the widely used CNN architecture, ResNet50 (He et al., 2016), into a novel Multi-level Deep Residual Learning (MDRL) system, concentrating on high-level and low-level semantic representations with a spatial attention mechanism. To better describe our method, we summarize the symbols in Table 3. ResNet50 is a capable, unified backbone network with a series of convolution blocks, and the classification task always follows several fully connected layers to transform the images into probabilities belonging to each category. This paper only employs the residual blocks in ResNet50, which is responsible for computing convolutional feature maps over an entire OCT image. In detail, the l -th residual block can produce a spatial feature map B_l ($1 \leq l \leq L$), where L is the number of residual blocks. As usual, the feature map B_L from the last residual block contains high-level semantic information for the OCT image but is absent for expressing detailed low-level information (He et al., 2020). That obstructs the subtle distinctions between malignant and benign oral tumors and neglects the essential features in the spatial feature map.

To overcome these problems and learn capable feature representations from OCT images, our proposed AI system integrates the high-level and low-level feature maps $\{B_1, \dots, B_l, \dots, B_L\}$ from all residual blocks into corresponding up-sample deconvolution layers, obtaining L corresponding feature hierarchy $\{H_1, \dots, H_l, \dots, H_L\}$ in a top-down pathway. This way strengthens the semantic information of the feature maps from high-level and low-level residual

blocks in a pyramid structure. Consequently, our MDRL system concatenates the series of pyramidal features into an entirety $H = [H_1, \dots, H_l, \dots, H_L]$ as the multi-level feature representation for each OCT image. Besides, the AI system develops a spatial attention mechanism on the last pyramidal feature H_L to further exploit the semantic information in high-level features. In detail, it is implemented by (1) spatial attentions $\{A^s\}$ to discover essential areas, and (2) channel attentions $\{A^c\}$ to integrate channel relation and send partial characteristics to the high-level pyramidal feature H_L . To obtain the optical weight parameters of our deep learning system, the diagnosing task (cross-entropy loss) is chosen on the concatenated pyramidal feature H and the attentive high-level feature $H_L \cdot (A^s \oplus A^c)$, optimized by the back-propagation algorithm. Through the optimization, the AI system accurately diagnoses OSCC with noninvasive OCT images, and the unified framework of MDRL is illustrated in Figure 1.

2.4 | Statistical analysis

The two-by-two statistical matrix is formulated by True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN), to compute the different measurements, such as sensitivity, specificity, accuracy, and the Receiver Operating Characteristic (ROC) curve (Zweig & Campbell, 1993), in the diagnostic validation. The specific calculations of sensitivity, specificity, accuracy, Positive Predictive Value (PPV), and Negative Predictive Value (NPV), false-positive, and false-negative rate of the diagnostic model to identify OSCC are computed as

$$\text{Sensitivity}(\%) = \frac{TP}{TP + FN} \times 100\% \quad (1)$$

$$\text{Specificity}(\%) = \frac{TN}{TN + FP} \times 100\% \quad (2)$$

$$\text{Accuracy}(\%) = \frac{TP + TN}{TP + FP + TN + FN} \times 100\% \quad (3)$$

$$\text{PPV}(\%) = \frac{TP}{TP + FP} \times 100\% \quad (4)$$

$$\text{NPV}(\%) = \frac{TN}{TN + FN} \times 100\% \quad (5)$$

TABLE 3 Summation of symbols

Symbol	Description	Symbol	Description
B_l	l -th spatial feature map	L	Number of blocks
H_l	Feature in l th level	H	Multi-level feature
A^s	Spatial attention	A^c	Channel attention
H'	Transformation of H	v_c	The parameter in deconvolution layer
H'_L	Attentive feature vector	y	Professional diagnosis (Label)

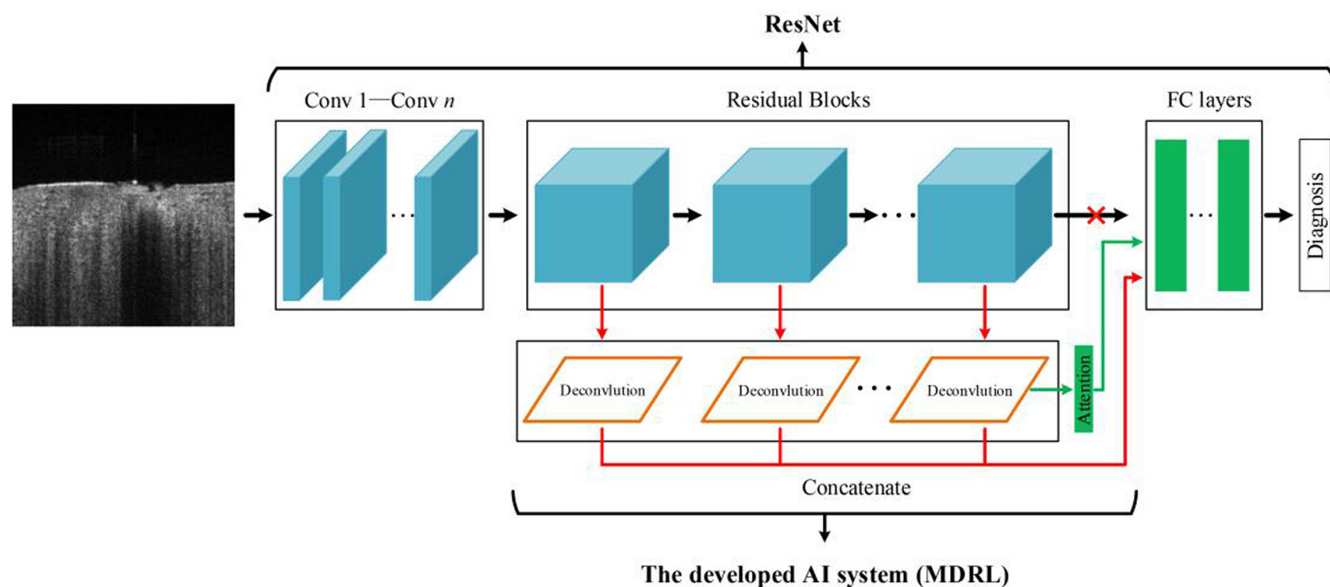


FIGURE 1 The framework of our proposed AI system for OSCC diagnosis with noninvasive OCT images. This system contains a backbone network of ResNet50 and involves deconvolution layers to generate pyramidal layers to obtain the multi-level pyramidal feature containing high-level semantic information and detailed low-level characteristics. Besides, we also introduce a spatial attention mechanism to enhance the semantic information for the last pyramidal features in the diagnosing task

Classifier	Sensitivity (%)	Specificity (%)	Accuracy (%)	PPV (%)	NPV (%)
Image-level diagnosis					
VggNet	86.3±5.3	72.1±4.5	78.6±3.2	72.1±4.9	86.3±4.4
GoogLeNet	89.1±4.9	79.0±4.2	83.9±3.8	80.3±5.2	88.2±4.2
ResNet	89.3±4.5	80.4±4.7	84.8±3.3	82.0±4.5	88.2±4.5
Our MDRL	91.2±3.7	83.6±3.6	87.5±1.9	85.3±2.6	90.2±3.8
Patch-level diagnosis					
Specialist	100	69.3	86.2	80.0	100
VggNet	100±0	81.3±7.8	89.7±4.3	81.3±8.2	100±0
GoogLeNet	86.7±5.8	78.6±7.5	82.8±4.0	81.3±7.8	84.6±4.2
ResNet	92.7±5.9	80.0±7.2	86.2±3.6	81.3±8.3	92.3±4.5
Our MDRL	100±0	86.7±6.8	93.1±2.1	87.5±6.5	100±0

TABLE 4 Comparison of diagnostic performance between the specialist, CNNs, and our MDRL

$$\text{False - positive rate}(\%) = 100 - \text{Specificity}(\%) \times 100\% \quad (6)$$

$$\text{False - negative rate}(\%) = 100 - \text{Sensitivity}(\%) \times 100\% \quad (7)$$

TP (True Positive): the number of images in which the AI model correctly identifies oral squamous cell carcinoma; FP (False Positive): the number of images in which the AI model falsely identifies OSCC; TN (True Negative): the number of images in which the AI model correctly identifies normal issues; FN (False Negative): the number of images in which the AI model falsely identifies normal issues.

Due to the OCT images being captured from different views of oral tissues, this work evaluates the diagnostic performance not only

at image-level but also at the resected patch-level. In addition, we also introduce the commonly utilized visualization methods to show the effectiveness of our AI model, such as the Class Activation Map (CAM) and loss curve.

3 | RESULTS

This study reports the diagnostic performance at the image-level and the resected patch-level in Table 4. We compare our AI system with three popular Convolutional Neural Networks (CNNs), and a specialist is also introduced to compare the performance in the resected patch-level.

3.1 | Comparison of diagnostic performances in image-level for diagnosing oral squamous cell carcinoma

The upper of Table 4 presents the diagnostic performances in the image-level, we compare our AI system (MDRL) with three traditional CNNs (VggNet (Simonyan & Zisserman, 2015), GoogLeNet (Szegedy et al., 2015), and ResNet (He et al., 2016)) for diagnosing OSCC in OCT images. In detail, our MDRL system achieves 91.2% sensitivity, 83.6% specificity, 87.5% accuracy, 85.3% PPV, and 90.2% NPV in image-level diagnosis, which is the highest diagnostic performance among different methods. Besides, the second-best results of 89.3% sensitivity, 80.4% specificity, 84.8% accuracy, 82.0% PPV, and 88.2 NPV% are performed by CNN3(ResNet), showing that our MDRL realizes a significant improvement over other CNNs, by integrating multi-level features and spatial attention mechanism.

To further demonstrate the effectiveness of our AI system, we also visualize the performance of different CNN models by the Receiver Operating Characteristic (ROC) curves in Figure 2. ROC analysis is a helpful tool for evaluating the performance of diagnostic tests and, more generally, for evaluating the accuracy of a binary classification model, for example, malignant or benign (Zou et al., 2007). Its function as a simple graphical tool for displaying the accuracy of a medical diagnostic test is one of the most well-known applications of ROC curve analysis. From Figure 2, we draw the ROC curves of our MDRL and three compared CNNs, marking their Area Under the Curve (AUC) values. It can be observed that Vgg, ResNet, GoogLeNet obtain 0.868, 0.895, and 0.917 AUCs, while our MDRL is the highest of 0.920. Thus, the excellent diagnostic performance of our MDRL AI system can be concluded, with a dominant AUC of 0.920.

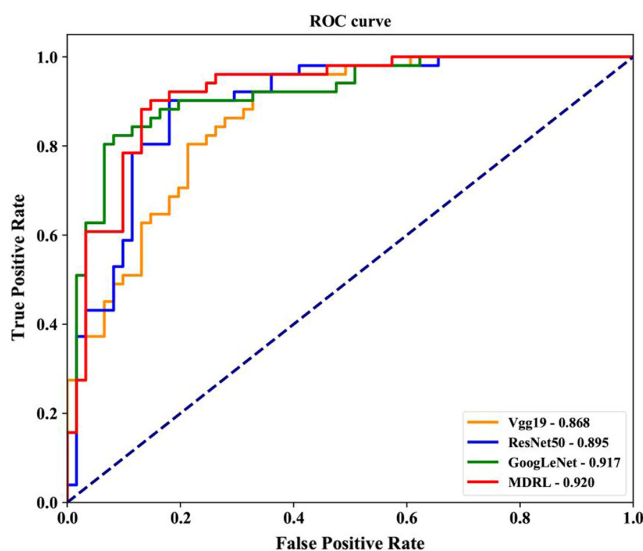


FIGURE 2 Comparison of the receiver operating characteristic (ROC) curves between CNNs and our AI model in the image-level diagnosis

We also make our deep learning system reveal the visual explanation for decisions of the OCT diagnosis by employing Gradient-weighted Class Activation Map (Grad-CAM) (Selvaraju et al., 2017). It uses the gradients of malignant and benign classes, flowing into the final convolutional layer to produce a coarse localization map highlighting the critical regions in the OCT image for predicting the malignant or benign. Using the Grad-CAM, we can visually validate where our system is looking, verifying that it looks at the correct patterns in the image and activates those patterns. Grad-CAM makes the CNN-based models more transparent by visualizing the regions of input that are “important” for predictions from the network. Specifically, the diagnosing experience of well-trained doctors is that the malignant OCT images always appear invisible basement membrane, no stratified structure of epithelium and stroma, disordered structure, and irregular refraction or cancer nest. From Figure 3, we can visualize the class activation maps of the proposed MRDL system, allowing us to verify that our network is “looking” and “activating” at the correct locations, where specialists observe the discriminative regions to make the diagnostic decision.

Moreover, the training cross-entropy loss of the diagnosing task reveals the performance with dynamic model optimization. The loss curve, along with training epochs, is visualized in Figure 4. The loss of our AI system is initially 0.14 at the beginning of the training, and it is rapidly decreased to around 0.02 after the first epoch. It becomes convergence after 10 epochs, with subtle fluctuations, and keeps stable in the last training process. The training tendency shows that our network rapidly becomes convergence and sufficiently demonstrates the training efficiency of our deep learning AI system.

3.2 | Comparison of diagnostic performances in the resected patch-level for diagnosing oral squamous cell carcinoma

In clinical practice, the doctors always conduct OCT imaging from multiple views for a resected patch of tissue. Here, our AI system predicts each view of the resected patch, computes the maximum predicted results for the multiple images from the same resected patch, and then outputs the resected patch-level diagnosis according to the maximum image-level results. The results are summarized in Table 4, and a manually predicted result from a specialist is also compared with our AI system and the CNNs. It can be easily seen that the resected patch-level diagnosis is preciser than image-level, with different improvements for each CNN method. As elaborated in Table 4, our MDRL AI system performs 100% sensitivity, 86.7% specificity, 93.1% accuracy, 87.5% PPV, and 100% NPV in the resected patch-level diagnosis. Compared with the specialist, our method has dominant superiority on specificity (17.4%), accuracy (6.9%), and PPV (7.5%), with equal diagnosing capability on sensitivity and NPV. This comparison demonstrates the diagnostic

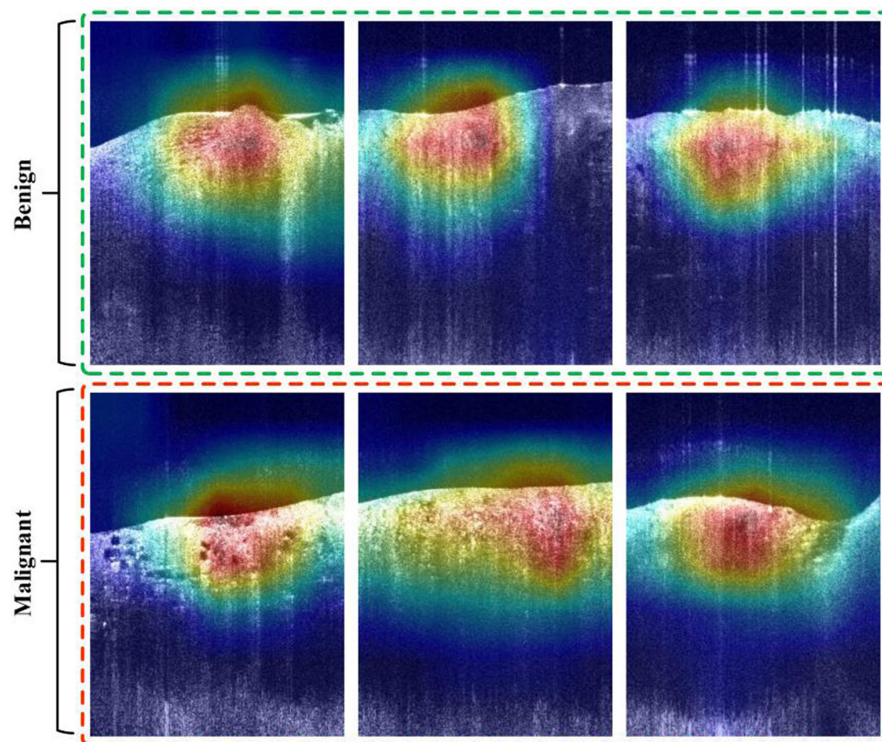


FIGURE 3 The gradient-weighted class activation map (grad-CAM) visualization for evaluation of our AI system

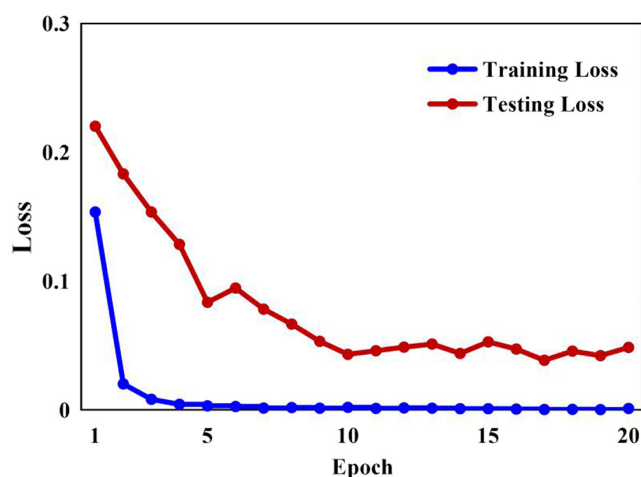


FIGURE 4 The training and testing loss curves with training epochs for evaluation of our AI system. The loss presents the fit between a prediction and the ground truth label. It decreases over time as training improves and will be stopped when the fluctuation between two training epochs is smaller than 5×10^{-5}

effectiveness of our AI system in the resected patch-level and reveals the superiority over the specialist and other CNNs.

4 | DISCUSSION

Several kinds of research have exploited the application of deep learning in automatic oral cancer diagnosis, which are mainly divided into two categories, including photographic image-based (Fu et al., 2020; Tanriver et al., 2021), and histological image-based

(Folmsbee et al., 2018; Lu et al., 2020). The photographic image-based methods often utilized deep learning to mine representative characteristics from photographic images of cavity squamous cell carcinoma patients. For example, Fu et al (Fu et al., 2020) developed a cascaded convolutional neural network to detect oral squamous cell carcinoma from photographic images, achieving AUC of 0.983, sensitivity of 94.9%, and specificity of 88.7% on 1469 intraoral photographic images; Tanriver et al (Tanriver et al., 2021) explored the potential applications of computer vision and deep learning techniques in the oral lesion domain within the scope of photographic images and investigated the prospects of an automated system for identifying oral potentially malignant disorders with two-stage pipeline. Another solution is to exploit information from histological images. Specifically, Lu et al (Lu et al., 2020) proposed a fully automated pipeline for oral squamous cell carcinoma detection on whole slide cytology images, and Folmsbee et al (Folmsbee et al., 2018) explored methods for efficiency training convolutional neural networks for tissue classification using active learning on the digital images of hematoxylin and eosin-stained whole oral cavity cancer sections. However, the photographic image-based models are sensitive to the cavity environments, such as occlusion, illumination, and the histological image-based methods require pathology resection, which is painful to patients and time-consuming for surgeons due to the complicated staining operations, resection, and imaging. Both types tend to integrate traditional CNN networks, only considering high-level semantic information to implement the diagnostic task and ignoring the different weights of the spatial features.

Aiming to establish an automated AI-based noninvasive oral squamous cell carcinoma diagnosis system, this study develops the

deep learning technology on Optical Coherence Tomography images, named Multi-level Deep Residual Learning model, solving the weakness of existing deep learning models mentioned before. As illustrated in Section 2.3, the high-level and low-level features can be obtained by L residual blocks in ResNet50. The residual blocks are skip-connection blocks that learn residual functions with reference that layer inputs instead of learning unreferenced functions, illustrated in Figure 5.

Through the up-sample devolution layers, each feature map from residual blocks can be transformed by $B_l \rightarrow H_l (l = 1, \dots, L)$, and then integrated into a multi-level pyramidal feature maps by concatenation $H = [H_1, \dots, H_L]$. Finally, the multi-level feature maps are transformed into a vector H' by Global Average Pooling (GAP),

$$H' = \text{GAP}(H) = \frac{1}{W \times H} \sum_{i=1}^H \sum_{j=1}^W H(i, j) \quad (8)$$

where W and H denote the width and height of the feature map.

Another improvement of our MDRL system is to design a spatial attention mechanism to enhance feature learning with high semantic information. As for the high-level feature vector H_L , it contains much more semantic representations for the malignant and benign tumors, and we firstly design a spatial attention A^s achieved by 3×3 deconvolution layer and a sigmoid function,

$$A^s = \sigma(v_c * H_L) \quad (9)$$

where σ denotes the sigmoid function, and v_c is the parameters in the deconvolution (*) layer. The spatial attention A^s can explore the different contributions of each element in high-level feature map H_L , and we also introduce a channel attention A^c to emphasize the importance of H_L in the channel-wise pathway, obtained by the GAP layer and two fully connected layers,

$$A^c = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(H_L))) \quad (10)$$

where W_1 and W_2 are the trainable parameters in two fully connected layers. Finally, the attentive feature vector can be calculated by,

$$H'_L = H_L \cdot (A^s \oplus A^c) \quad (11)$$

where $\oplus$ indicates the addition operation.

After obtaining the multi-level feature vector H' and high-level attentive feature H'_L , the prediction loss is integrated to ensure the system can determine a correct diagnosis of malignant or benign tumor. The prediction loss is defined by

$$\mathcal{L} = - \left(\sum_x y \log(p(H')) + \sum_x y \log(p(H'_L)) \right) \quad (12)$$

where x denotes the input OCT image of our network, and y is the professional diagnosis for x .

Benefiting from the multi-level feature integration and high-level attention mechanism, our AI system is implemented on 460 OCT images from 37 patients with primary oral squamous cell carcinoma in Beijing Stomatological Hospital, Capital Medical University, China. The system achieves the excellent and superior diagnosis performance on OCT images, with 91.2% sensitivity, 83.6% specificity, 87.5% accuracy, 85.3% PPV, and 90.2% NPV in image-level, and 100% sensitivity, 86.7% specificity, 93.1% accuracy, 87.5% PPV, and 100% NPV in the resected patch-level. This study demonstrates that deep learning technology has potential in noninvasive OSCC diagnosis, and its performance can be largely improved by introducing multi-level feature learning and spatial attention mechanism.

In this study, we randomly divided the OCT dataset into training and testing sets in the resected patch-level, which may contain mixed patients in training and testing data. However, only a few patients are mixed, and the resected patches from the same patient have sufficient distribution difference because their OCT images are captured independently with different conditions, such as illumination, viewpoint, and benign or malignant. Thus, the mixed patients in image patches have a limited impact on the generality of our proposed MDRL model, and this study can be easily applied to patient-level OSCC diagnosis.

In the future, we will further develop the AI system to diagnose OSCC in noninvasive OCT images with higher accuracy, with exploitation by the 3D convolutional neural network. Besides the prediction of malignant or benign OCT images, another critical issue in the clinical treatment of oral squamous cell carcinoma is to detect the edge of the tumor, which can help the surgeons to resect the cancer tissue with a noninvasive examination, due to the OCT is a

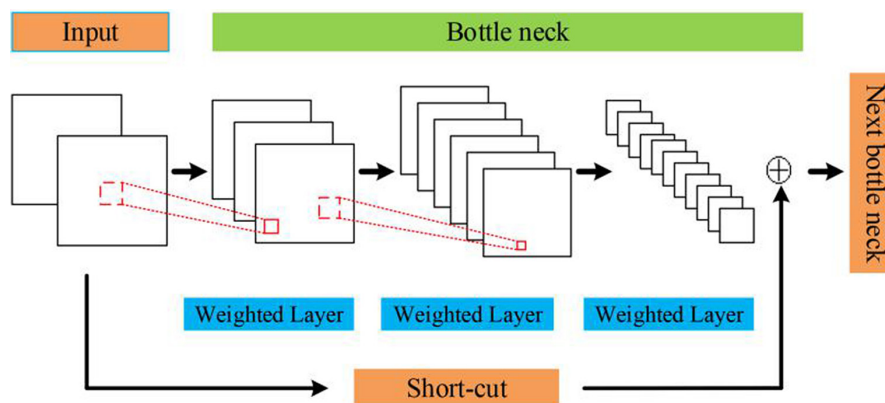


FIGURE 5 The architecture of a residual block in ResNet

real-time, invasive, and efficient imaging technology. The main challenge in future is how to annotate sufficient OCT images to support the model training, and we will also study the semi-supervised and unsupervised learning approaches to solve this problem. In conclusion, this study opens a new window for the computer-aided diagnosis system of diagnosing, detecting, and segmentation for OSCC in a noninvasive manner.

AUTHOR CONTRIBUTIONS

Wei Yuan: Conceptualization; data curation; formal analysis; investigation; methodology; resources; validation; writing – original draft; writing – review and editing. **Jinsuo Yang:** Data curation; formal analysis; investigation; visualization; writing – original draft; writing – review and editing. **Boya Yin:** Data curation; formal analysis; validation; writing – original draft. **Xingyu Fan:** Data curation; investigation; resources; validation. **Jing Yang:** Data curation; formal analysis; resources; validation. **Haibin Sun:** Data curation; formal analysis; investigation; resources. **Yanbin Liu:** Data curation; formal analysis; investigation; resources. **Ming Su:** Formal analysis; writing – original draft. **Sen Li:** Formal analysis; validation. **Xin Huang:** Conceptualization; formal analysis; funding acquisition; investigation; project administration; supervision; writing – original draft; writing – review and editing.

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CONFLICT OF INTEREST

We declare that we have no conflict of interest. Informed consent was obtained from all individual participants included in the study.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Wei Yuan  <https://orcid.org/0000-0001-5772-6847>

Xin Huang  <https://orcid.org/0000-0002-6648-6678>

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